Fiscal Year:	FY 2016	**Task Last Updated:**	FY 03/29/2016
PI Name:	De Witt, John Ph.D.		
Project Title:	Retrospective Analysis of Inflight Exercise Loading and Health Outcomes		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77058	**Congressional District:**	22
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	03/07/2014	**End Date:**	03/30/2016
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
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Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Ploutz-Snyder, Lori Ph.D. (Universities Space Research Association/NASA Johnson Space Center)		
Grant/Contract No.:	Directed Research		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Astronauts perform exercise throughout their missions to counter the health declines that occur as a result of long-term exposure to weightlessness. The current exercise devices on the International Space Station (ISS) allow astronauts to complete prescriptions at higher intensities, potentially allow for greater exercise benefits with increased efficiency. However, although overall health outcomes have improved, the specific factors related to increased benefits have yet to be determined. The ability of current exercise devices to collect performance data allows for exploratory analyses to be performed determining if specific factors such as external loading are related to exercise outcomes. These analyses are vital for understanding which components of exercise are most critical for program success. This retrospective analysis will be conducted using data from seven crewmembers that were subjects in the Treadmill Kinematics study. Inflight ground reaction force data from these subjects indicates that subject-specific differences exist in foot forces generated when exercising on the T2. The relationship between these differences and health changes has yet to be investigated. The information generated from this analysis will fill gaps related to typical bone loading characterization, exercise		

	performance capability, exercise volume and efficiency, and importance of exercise hardware. Specific Aims: 1) To quantify the exercise-related mechanical loading experienced by crewmembers during their mission on ISS; 2) To explore relationships between exercise loading variables, bone, and muscle health changes during the mission; 3) To determine if specific mechanical loading variables are more critical than others in protecting health; 4) To develop methodology for operational use in monitoring crew exercise programs.
Rationale for HRP Directed Research:	This research is directed because it contains highly constrained research, which requires focused and constrained data gathering and analysis that is more appropriately obtained through a non-competitive proposal
Research Impact/Earth Benefits:	N/A
Task Progress:	Astronauts perform exercise throughout their missions to counter space-related health declines. The ability of the T2 to collect performance data allows for the performance of exploratory analyses to find if specific factors, such as prescription or external loading, are related to exercise outcomes. These analyses are vital for understanding which components of exercise are most critical for exercise program success. This retrospective analysis was conducted using data from seven crewmembers that were subjects in the Treadmill Kinematics study. Exercise logs and inflight ground reaction force data were summarized to quantify T2 mechanical loading occurring during flight. Mechanical loading variables were compared to bone mineral density changes to determine if relationships existed. Results indicated that increased time and distance jogging related to improved bone health. Further investigation revealed that in the current subject pool, those who performed exercise sessions that include both running and jogging, such as during high intensity interval training, tended to have better bone-related outcomes. More study is necessary, including increasing subject size and inclusion of resistance exercise to better assess exercise relationships to bone health.
Bibliography Type:	Description: (Last Updated: 02/11/2021)
Articles in Peer-reviewed Journals	De Witt JK, Ploutz-Snyder LL. "Ground reaction forces during treadmill running in microgravity." Journal of Biomechanics. 2014 Jul 18;47(10):2339-47. Epub 2014 Apr 30. http://dx.doi.org/10.1016/j.jbiomech.2014.04.034 ; PubMed PMID: 24835563 , Jul-2014
Articles in Peer-reviewed Journals	De Witt JK, Edwards WB, Scott-Pandorf MM, Norcross JR, Gernhardt ML. "The preferred walk to run transition speed in actual lunar gravity." J Exp Biol. 2014 Sep 15;217(Pt 18):3200-3. http://dx.doi.org/10.1242/jeb.105684 ; PubMed PMID: 25232195 , Sep-2014